AASTMT · NAUTICAL DEPARTMENT

MODEL ANSWER

Introduction to Navigation (BS131) — 7th Week Exam

06 April 2025 | **1 Hour** | **20 Marks**

Final answers are highlighted in green. Short workings are shown only where needed.

Question 1 — Definitions & Positions

7 Marks

A · COMPLETE THE SENTENCES

1-a. Parallel of latitude … parallel to the plane of …….

Plane of the parallel is parallel to the equatorial plane.

ANSWER the Equator

1-b. Rhumb line … cuts all ……. at the same angle.

Constant course ⇒ same angle with every meridian.

ANSWER Meridians

1-c. Meridians are semi great circles joining …….

Each meridian joins the poles and is perpendicular to the equator.

ANSWER the North and South Poles

B · POSITION CALCULATIONS

B-1. A: 28° 30.8′ S, 057° 28.6′ W → B: 29° 42.6′ N, 069° 33.4′ W

```
D.Lat = 29° 42.6′ – (–28° 30.8′) = 58° 13.4′ N
D.Long = 069° 33.4′ W – 057° 28.6′ W = 12° 04.8′ W
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ANSWER D.Lat = 58° 13.4′ N | D.Long = 12° 04.8′ W

B-2. Start 33° 24.8′ N, 172° 15.6′ E · D.Lat 22° 56.7′ S · D.Long 011° 57.8′ E

```
Lat = 33° 24.8′ N – 22° 56.7′ = 10° 28.1′ N
Long = 172° 15.6′ E + 011° 57.8′ E = 184° 13.4′ E = 175° 46.6′ W
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ANSWER 10° 28.1′ N , 175° 46.6′ W

Question 2 — Figure Extraction & Gnomonic Chart

6 Marks

A · FROM THE FIGURE

M.N = 6° W of T.N · C.N = 2° E of T.N · Heading from C.N = 085° · Lighthouse from T.N = 122°

Variation

ANSWER 6° W

Total Compass Error

ANSWER 2° E

Compass Course

ANSWER 085° C

True Course

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085° + 2° E = 087° T
```

ANSWER 087° T

True Bearing

ANSWER 122° T

Relative Bearing

```
122° – 087° = 035° Green
```

ANSWER 035° Green

B · GNOMONIC TERMS (Seattle → London)

ANSWER 1. Initial Course · 2. Vertex · 3. Final Course

Question 3 — Compass, Bearings & Gyro

7 Marks

A · FILL THE TABLE

Rule: Mag = Comp ± Dev (East add · West subtract)

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1) Mag = 155° – 3° W = 152° M
2) Comp = 273° – 5° E = 268° C
```

	C.Course	Dev.	Mag.Course
1	155°	3° W	**152° M**
2	**268° C**	5° E	273°

B · BEARINGS

B-1. T.Co 023° · Rel 045° Red

```
023° – 045° = –022° → 338° T
```

ANSWER 338° T

B-2. T.Co 111° · T.Bg 140°

```
140° – 111° = 029° Green
```

ANSWER 029° Green

C · ERRORS & CONVERTED BEARINGS

True 333° T · Compass 322° C · Gyro 335° · Dev 5° E

1. Total Compass Error

```
333° – 322° = 11° E
```

ANSWER 11° E

2. Variation

```
C.E. – Dev = 11° – 5° = 6° E
```

ANSWER 6° E

3. Gyro Error	4. T.Bg 359° → Comp & Gyro
`335° – 333° = 2° High`	`359–11=348° C · 359+2=001° G`
ANSWER 2° High	**ANSWER** 348° C | 001° G

Quick Sheet — Final Answers Only

Summary

Item	Final Answer
Q1-A	Equator / Meridians / North & South Poles
Q1-B1	D.Lat 58° 13.4′ N · D.Long 12° 04.8′ W
Q1-B2	10° 28.1′ N , 175° 46.6′ W
Q2-A	Var 6°W · CE 2°E · C.Co 085°C · T.Co 087°T · T.Bg 122°T · Rel 035°G
Q2-B	1 Initial Course · 2 Vertex · 3 Final Course
Q3-A	Mag 152° M · Comp 268° C
Q3-B	T.Bg 338° T · Rel 029° Green
Q3-C	CE 11°E · Var 6°E · GE 2° High · Comp.Bg 348°C · Gyro.Bg 001°G

Cadet (Compass Add East True) · Error East / Compass Least · Red = Port · Green = Starboard · Gyro High/Low